PROJECT ACTION PLAN

MailGuard

An email deliverability diagnostic tool built for non-technical African SME business owners

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Status: Pre-build / Planning Phase

Target MVP timeline: 4–6 weeks

Pace: Side-project (a few hours/week)

1. The Problem

> Non-technical African SME owners can't fix email deliverability issues because existing tools (Mail-Tester, MXToolbox, GlockApps) are built for developers, not for them — leaving the actual problem unsolved.

Success Definition

"Success is a business owner pasting their email, getting a clear answer in under 2 minutes on why it's landing in spam, following simple steps to fix it, and confirming — via re-test — that it now passes."

2. Core Objectives

1. **Diagnose email deliverability issues automatically** — check SPF, DKIM, DMARC configuration, spam score, and blacklist status.
2. **Explain results in plain, non-technical language** — translate technical findings into clear, actionable explanations using an AI layer.
3. **Provide actionable fix instructions** — tell the user exactly what to change, where, and how.
4. **Allow re-verification after fixes** — a simple re-test button to confirm the issue is resolved.
5. **Keep the platform secure and trustworthy** — proper auth, encryption, and data retention limits.
6. **Make it accessible and affordable for African SMEs** — simple UI, clear language, fair pricing — the core differentiator from existing tools.

3. The Action Plan

Phase 0 — Alignment

Deadline: This week **Dependency:** None — must happen first

Call with friend, walk through the concept, lock roles, agree on stack.

Phase 1 — Foundation Setup

Deadline: End of Week 1 **Dependency:** Phase 0 complete

Set up repo, project structure, README, PostgreSQL schema (users, scans, reports), Express API skeleton + JWT auth (you), and SpamAssassin Docker container (friend).

Phase 2 — Core Diagnostic Engine

Deadline: End of Week 3 **Dependency:** Phase 1 complete

Build DNS lookup logic for SPF/DKIM/DMARC (friend), connect SpamAssassin scoring to backend (friend), build input form (you), build endpoint that triggers analysis (shared).

Phase 3 — AI Explanation Layer

Deadline: End of Week 4 **Dependency:** Raw diagnostic results from Phase 2 working

Integrate Claude API to translate technical results into plain English; build the color-coded results/report UI.

Phase 4 — Fix Instructions + Re-test

Deadline: End of Week 5 **Dependency:** Phases 2 and 3 fully working

Generate step-by-step fix instructions per issue type; build the "Re-test" button and re-run logic.

Phase 5 — Security & Polish

Deadline: End of Week 6 **Dependency:** All previous phases working

Rate limiting, input sanitization, data retention/privacy cleanup, error handling, mobile responsiveness, end-to-end testing with real emails (e.g. Zora Market).

Phase 6 — Soft Launch / Feedback

Deadline: Ongoing after Week 6 **Dependency:** Stable MVP

Share with a handful of real business owners, collect feedback, fix critical bugs.

4. Required Resources

Budget (MVP phase)

Item	Estimated Cost
Domain name	~₦5,000–15,000/year
Frontend hosting (Vercel/Netlify free tier)	₦0
Backend hosting (Railway/Render)	₦0–₦7,000/month
PostgreSQL (Railway included, or Supabase free tier)	₦0
SpamAssassin (open source, self-hosted)	₦0

Claude API (pay-as-you-go, low volume testing)	~$5–20/month
DNS lookup tools	₦0 (free public tools)
SSL/HTTPS	₦0 (Let's Encrypt / Vercel / Railway)

Realistic total: under $30–50/month during MVP/testing phase.

Tools / Tech Stack

- **Frontend:** React, Tailwind CSS
- **Backend:** Node.js, Express
- **Database:** PostgreSQL
- **Spam scoring engine:** Apache SpamAssassin (Dockerized)
- **DNS validation:** Node DNS libraries
- **AI explanation layer:** Claude API
- **Auth:** JWT
- **Version control:** Git + GitHub
- **Deployment:** Vercel (frontend), Railway/Render (backend)
- **Project tracking:** Trello, Notion, or GitHub Projects

People

Task	Owner
Frontend UI (forms, dashboard, results page)	You
Auth (JWT)	You
Backend API skeleton & routes	You
Claude API integration (explanation layer)	You
DNS / SPF / DKIM / DMARC validation logic	Friend
SpamAssassin setup & integration	Friend
Blacklist checking	Friend
Database schema design	Shared
Security (rate limiting, sanitization, data handling)	Shared
Deployment & DevOps	Shared
Testing with real emails	Shared

5. Potential Roadblocks & Mitigations

Risk	Plan B
Friend gets busy or drops off (school, internship, life)	Define clear, separable ownership early so you can ship a smaller version solo if needed.
SpamAssassin setup proves harder than expected	Fall back to Postmark's free Spam Score API — no server management needed.
DNS/SPF/DKIM/DMARC logic is more complex than anticipated	Start with read-only pass/fail checks before attempting auto-fix suggestions; use existing libraries.
Claude API costs scale unexpectedly	Set usage caps/budget alerts early; cache repeated explanations for common issues.
Scope creep (ESP sending, game features, ML v2 too early)	Keep the locked MVP scope visible daily: Diagnose → Explain → Fix → Re-test.
Real test emails are hard to source	Use your own Zora Market emails and Mailerlite account as the first real test case.
Security gaps in storing sensitive email content	Don't store full email content longer than needed; encrypt at rest; clear retention policy from day one.
Motivation dips mid-build (common in 4–6 week side projects)	Set small public milestones — share progress weekly for light accountability.
Time estimate slips past 6 weeks	Treat Week 6 as a soft deadline; Week 7–8 is an acceptable stretch before reassessing.

6. Metrics for Success

MVP-Stage KPIs (proves the product works)

- **Diagnosis accuracy** — correctly identifies real SPF/DKIM/DMARC issues against known broken setups (including your own Mailerlite case).
- **Time to result** — under 2 minutes from input to a readable answer.
- **Comprehension test** — a non-technical person (not you or your friend) can read the report and explain what's wrong in their own words.
- **Fix success rate** — of users who follow the instructions, how many pass the re-test afterward.

Early Traction KPIs (proves people want it — Week 6+)

- Number of real business owners who test it (target: 5–10 real people).
- Completion rate — of people who start a scan, how many finish and read the result.
- Return usage — do people come back to re-test after fixing an issue.

- Qualitative feedback — statements like "I finally understand why my emails were failing."

> **Avoid measuring (too early):** revenue, number of paying users, growth rate. Chasing these before the core loop works will distract from actually shipping.